

XIANYANG, China

WMO: 570360

Lat: 34.447N

Lon: 108.752E

Elev: 479

StdP: 95.70

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -7.0	(d) -18.2	(e) 0.8	(f) 0.3	(g) -16.0	(h) 1.0	(i) 0.1	(j) 7.5	(k) 2.3	(l) 6.6	(m) 1.1	(n) 1.2	(o) 270	(p) 0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	36.8	23.3	35.1	23.0	33.8	22.8	26.4	32.5	25.7	31.4	24.9	30.4	3.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	21.3	29.8	24.1	20.1	29.2	23.2	19.0	28.3	86.2	33.1	82.6	31.5	79.0	30.6	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.1	6.1		-11.6	39.3	3.0	1.1	-13.7	40.1	-15.5	40.8	-17.2	41.4	-19.4	42.2
			DB	-12.4	27.8	2.7	1.0	-14.4	28.5	-16.0	29.1	-17.5	29.6	-19.4	30.3
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	-0.1	3.8	9.7	15.7	20.6	25.5	27.5	25.8	20.6	14.5	7.4	1.5
	DBStd	10.00	3.12	3.57	3.97	3.85	3.25	3.12	2.95	3.16	3.40	3.28	3.55	2.86
	HDD10.0	906	314	173	54	4	0	0	0	0	0	4	92	265
	HDD18.3	2348	572	406	267	96	16	1	0	1	14	125	327	523
	CDD10.0	2521	0	1	46	175	328	464	542	489	318	143	14	0
	CDD18.3	920	0	0	0	17	86	215	284	231	82	6	0	0
	CDH23.3	9756	0	0	19	228	871	2451	3271	2259	603	55	0	0
	CDH26.7	4471	0	0	2	58	319	1240	1657	989	198	8	0	0
Wind	WSAvg	2.4	1.9	2.3	2.7	2.7	2.6	2.6	2.8	2.7	2.4	1.9	1.9	1.9
Precipitation	PrecAvg	529	6	9	23	36	56	61	86	84	89	51	23	6
	PrecMax	866	19	28	58	102	143	112	209	196	300	128	65	18
	PrecMin	295	0	0	0	6	3	6	5	9	15	10	0	0
	PrecStd	123	5	7	15	22	35	30	43	46	64	28	19	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	17.1	25.6	31.1	34.8	38.1	39.0	37.0	34.2	27.5	20.9	14.0
		MCWB	4.8	7.6	14.1	18.3	18.8	21.0	24.6	25.3	23.2	16.8	11.6	5.7
	2%	DB	9.7	14.8	22.8	28.1	31.9	36.2	37.0	35.0	31.1	24.9	18.0	11.0
		MCWB	2.9	6.7	12.5	17.0	18.5	21.1	24.3	24.9	21.8	16.0	10.9	4.2
	5%	DB	7.2	12.2	20.0	26.0	30.1	34.9	35.2	33.1	28.9	22.8	15.9	9.0
		MCWB	1.6	5.6	11.0	16.3	18.3	20.8	24.3	24.4	20.9	15.5	9.8	3.3
	10%	DB	5.7	10.2	17.2	23.8	28.2	33.0	33.9	31.8	26.9	20.7	13.9	7.1
		MCWB	0.8	4.5	9.7	15.1	18.0	20.4	23.9	24.1	20.0	14.7	9.1	2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	9.2	15.7	21.0	22.1	25.1	27.7	27.5	25.2	19.3	13.5	6.7
		MCDB	11.7	15.0	22.5	27.7	29.0	31.9	34.0	33.2	31.2	23.4	17.7	11.0
	2%	WB	3.4	7.5	13.3	18.7	20.9	23.9	26.7	26.5	23.5	18.0	12.0	5.1
		MCDB	8.2	12.8	20.6	25.5	27.8	31.0	33.1	32.1	28.6	21.6	15.4	9.4
	5%	WB	2.2	6.4	11.9	17.2	19.9	23.0	25.9	25.7	22.0	16.8	11.0	3.9
		MCDB	6.5	10.8	18.2	23.9	26.8	30.0	32.2	30.9	26.5	20.6	14.8	7.6
	10%	WB	1.1	5.3	10.4	15.9	19.1	22.0	25.2	24.9	20.9	15.7	9.8	2.9
		MCDB	4.9	9.5	16.4	22.0	25.6	29.4	31.1	29.9	25.0	19.2	13.0	6.2
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.2	11.4	12.1	12.1	12.0	10.0	9.1	8.9	9.5	9.8	9.8
		MCDBR	13.2	14.3	16.5	16.6	16.2	16.0	12.9	11.5	12.8	13.8	13.8	13.8
		MCWBR	8.1	8.3	8.6	7.8	6.5	5.2	4.4	4.4	5.4	6.5	7.7	8.3
	5% WB	MCDBR	11.7	12.5	14.9	14.3	12.9	11.5	10.3	9.8	10.0	10.7	11.6	12.0
		MCWBR	7.4	7.6	8.1	7.6	6.1	4.8	4.4	4.5	5.0	5.6	6.8	7.6
Clear-Sky Solar Irradiance	taub	0.541	0.621	0.637	0.629	0.612	0.638	0.661	0.698	0.652	0.630	0.573	0.509	
	taud	1.699	1.572	1.574	1.624	1.721	1.711	1.708	1.620	1.674	1.658	1.700	1.779	
	Ebn at Noon	612	611	654	693	713	693	673	637	637	597	579	611	
	Edh at Noon	189	239	258	256	235	237	236	254	230	216	188	165	
All-Sky Solar Radiation	RadAvg	2.52	2.97	4.06	4.83	5.18	5.37	5.14	4.46	3.48	3.00	2.57	2.37	
	RadStd	0.28	0.48	0.40	0.47	0.41	0.49	0.66	0.63	0.40	0.46	0.37	0.32	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		-0.40	-2.26	-1.39	N/A	N/A	N/A	+61	+79	N/A	-71
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page